

The Effects of Augmented Reality Head-Up Display Graphics on Driver Situation Awareness and Takeover Performance in Driving Automation Systems

Richard L Greatbatch¹
Virginia Tech

Hyungil Kim²
Oakland University

Zachary Doerzaph³
Virginia Tech Transportation Institute



Figure 1. Example of world-relative (left) and screen-relative (right) AR HUD graphic TORs

ABSTRACT

Implementing head-up displays to convey takeover requests allows for the opportunity to increase driver safety during takeover. This work aims to gain greater understanding in the effects of different visualization strategies on takeover performance and participant's situation awareness using two HUD visualization strategies, screen-relative and world-relative graphics. Initial results indicate an increase in takeover performance using HUD graphic visualization strategies.

Keywords: Driving Automation System, Takeover, Augmented Reality, HUD

Index Terms: H.5 [Information Interfaces and Presentation]: H.5.1: Multimedia Information Systems — Artificial, Augmented, and Virtual Realities; H.5.2: User Interfaces — Ergonomics, Evaluation / Methodology

1 INTRODUCTION

Various technologies exist utilizing SAE Level 3 automation. These systems require drivers to remain vigilant and fallback ready if system capabilities are exceeded [1]. In situations where the system exceeds its capabilities, drivers will receive a takeover request (TOR), indicating the need for drivers to regain vehicle operation. To safely regain operation, drivers must have a proper understanding of their surroundings, or situation awareness (SA) [2]. Currently, TORs utilize head-down displays (HDD) that take drivers eyes off the road to receive important visual information

[3]. Future human-machine interfaces may employ the use of novel technologies such as an augmented reality (AR) head-up display (HUD) to deliver visual information to drivers during takeover. Typically, HUDs utilize two visualization strategies, screen-relative and world-relative graphics. Screen-relative graphics deliver visual information to drivers while being attached to the HUD display on the windshield. While world-relative graphics deliver visual information spatially attached to the world itself [4]. However, world-relative graphics are not naturally better than screen-relative graphics in all situations, so it is important to understand effectiveness of different visualization strategies in takeover [5].

2 OBJECTIVE

Currently, TORs using AR HUDs face many challenges during takeover. This study sought to investigate the effects of AR HUDs and Head Down Display (HDDs) graphics on driver situation awareness (SA) and takeover performance.

3 METHOD

We conducted a user study involving 21 participants where participants operated a vehicle with a Level 3 driving automation system. Participants completed a series of 30 drives resulting in takeover. Each participant experienced a TOR from three visual graphic types, world-relative HUD (Figure 1 left), screen-relative HUD (Figure 1 right), and HDD. Upon completion of each trial, we measured driver SA and takeover performance.

4 RESULTS

We found that participant's overall SA scores did not significantly vary between the HDD, screen-relative HUD, and world-relative HUD graphic conditions. Mean scores for each graphic condition revealed that participants in the HDD graphic condition showed a

¹rlg1990@vt.edu

²hyungilkim@oakland.edu

³zdoerzaph@vti.vt.edu

mean overall SA score of 78.55%. Participants in the world-relative HUD graphic conditions showed average scores of 82.45% and screen-relative HUD average scores of 81.69%. Participant's initial response time (IRT) indicated higher IRTs were found between the HDD graphic condition when compared to the world-relative HUD graphics conditions. Where participants had an average ITRs of 1.97 seconds in the HDD graphic condition, 1.54 seconds in the world-relative HUD condition, and 1.70 seconds in the screen-relative HUD condition. Participant's time to collision (TTC) indicated that participants in the world-relative HUD graphic conditions had the greatest TTC. Mean scores between visual TOR cues show that participants had the highest average TTC of 2.71 seconds in the world-relative HUD graphic conditions, followed by 2.34 seconds in the screen-relative HUD graphic conditions, and 2.11 seconds in the HDD graphic conditions. Participants were nine times as likely to carry out the correct action in the world-relative graphic condition than the HDD graphic condition. Participants in the screen-relative HUD graphic condition were 3.5 times as likely to carry out the correct action than the HDD.

5 CONCLUSION

Results did not directly support the use of AR HUDs to increase driver SA during takeover in vehicles equipped with driving automation systems, but participants were shown to have increased takeover performance regarding initial response time, time to collision, and reaction accuracy.

6 APPLICATION

Results of this study found that HUDs support increased driver performance which may help inform future designs of TORs for Level 3 driving automation systems.

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